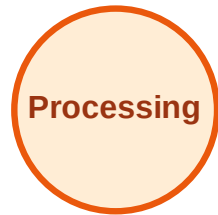


BFS Traversal

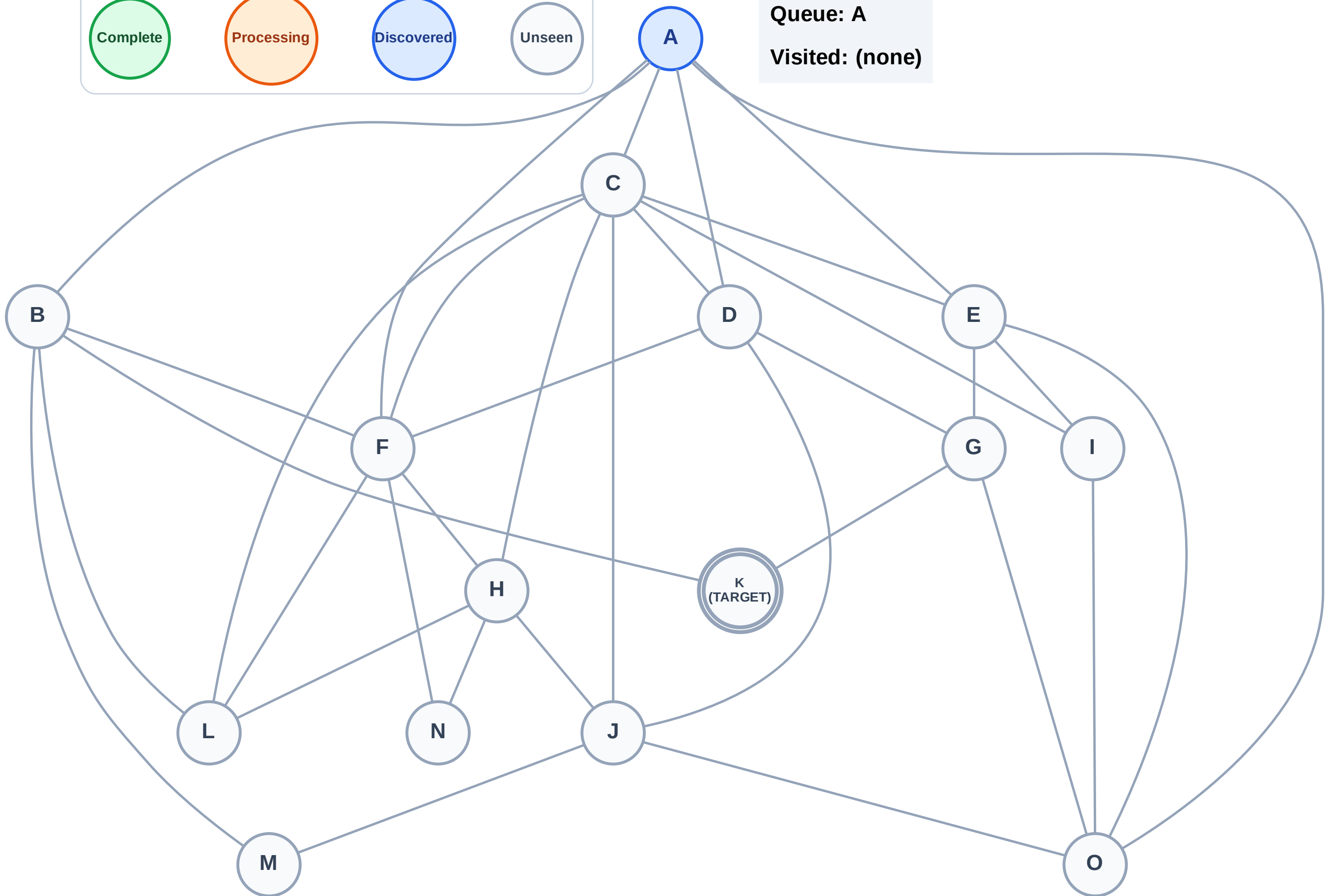
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

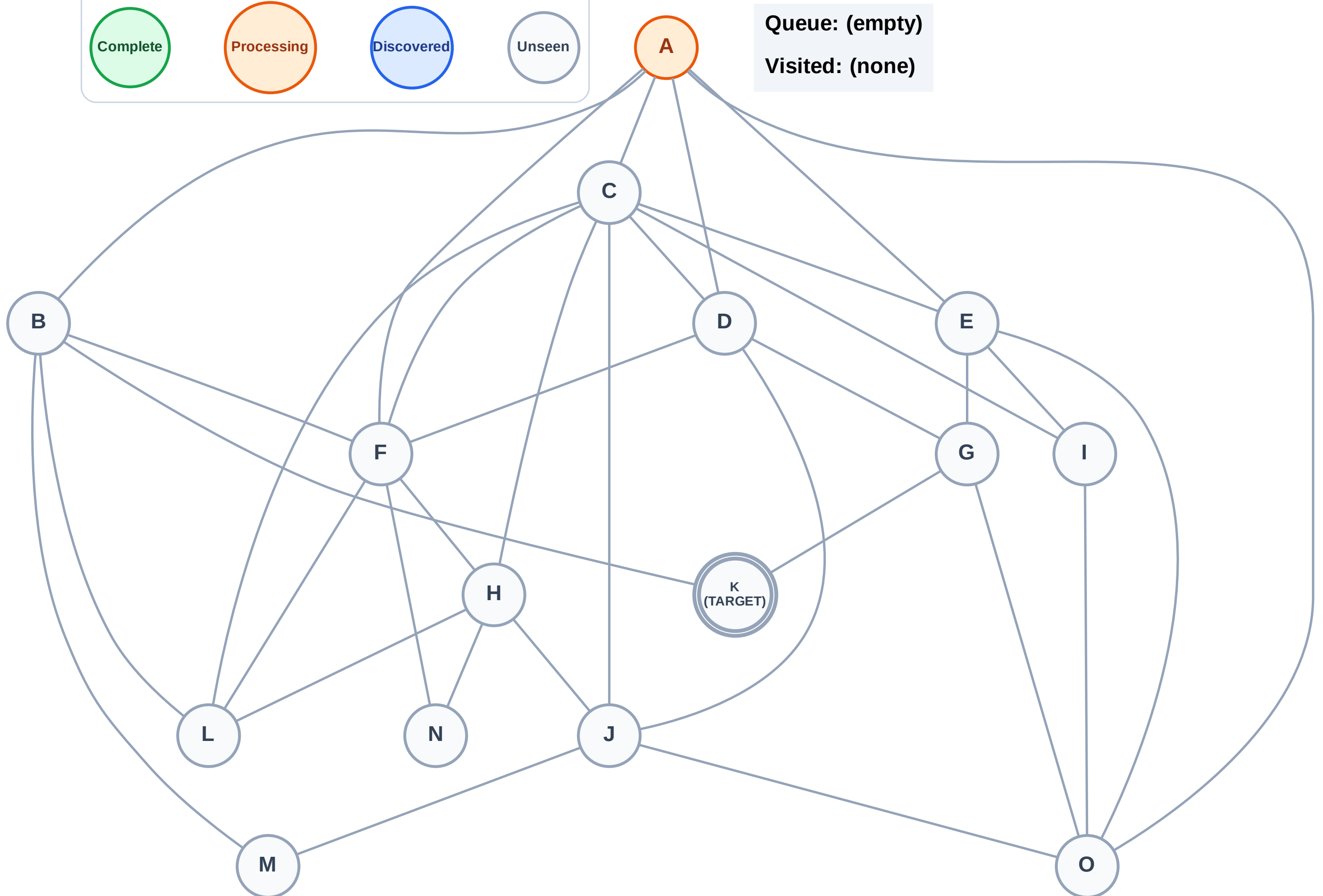
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

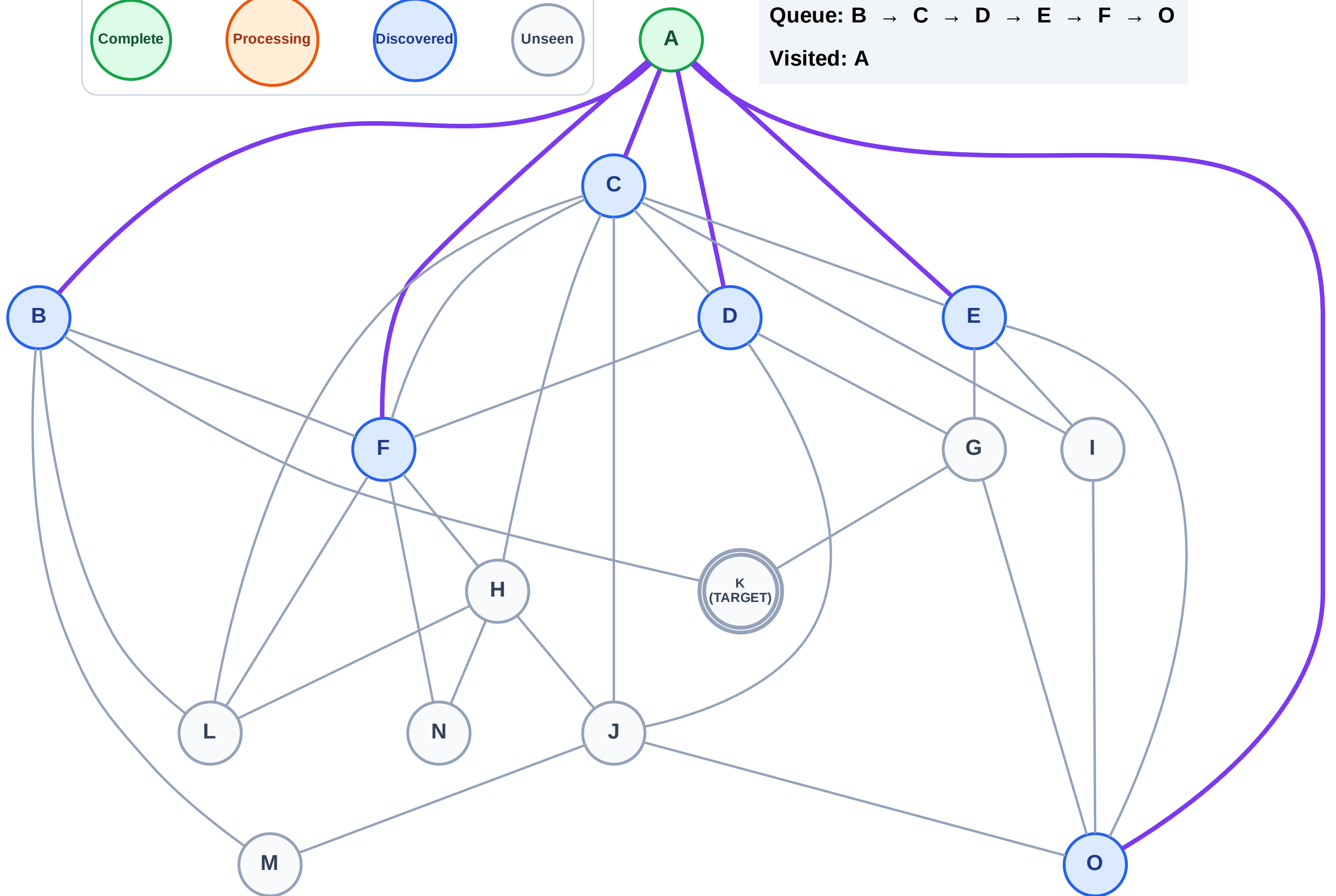
Step 3: Discover B, C, D, E, F, O from A

Legend



Queue: B → C → D → E → F → O

Visited: A



BFS Traversal

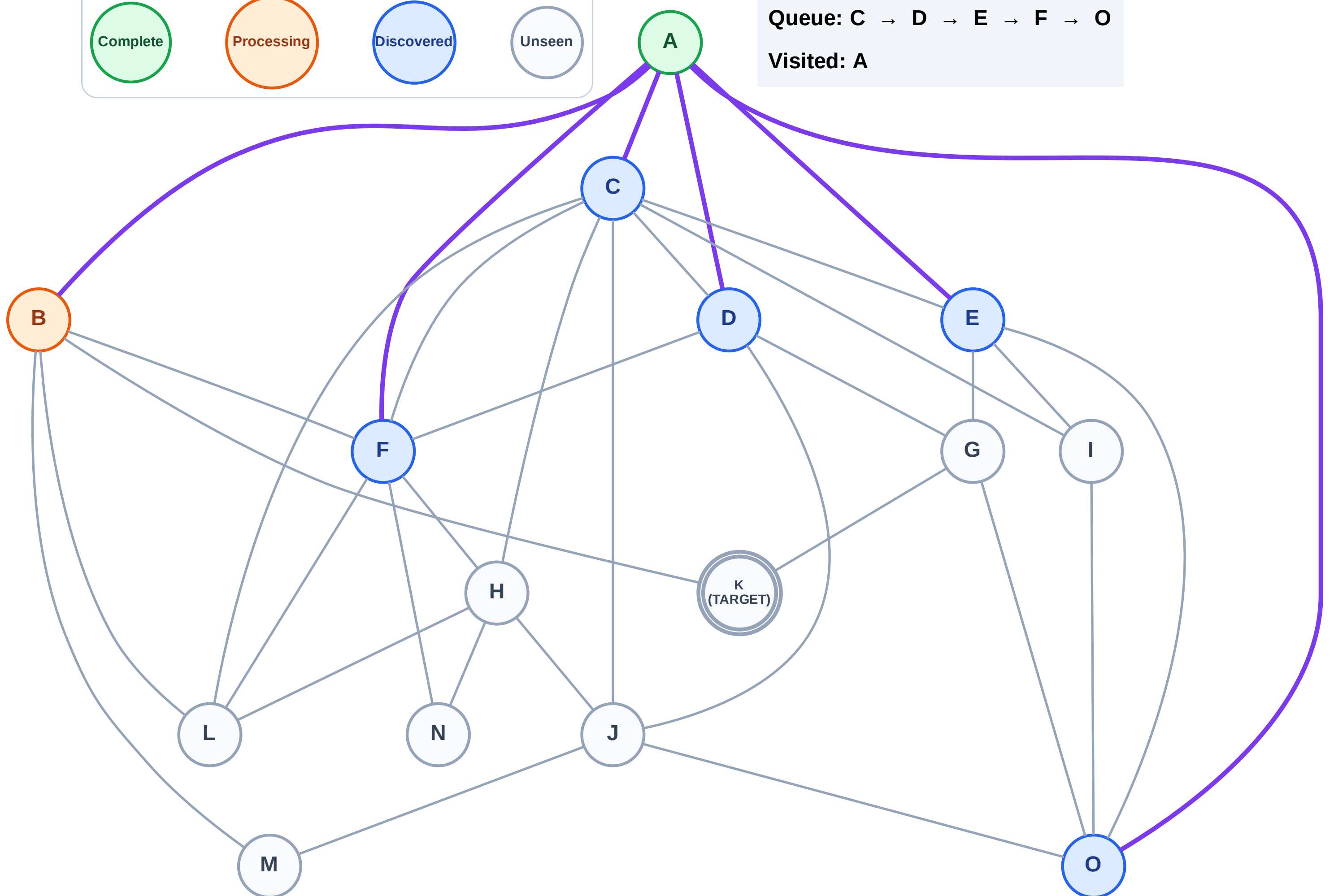
Step 4: Process node B

Legend



Queue: C → D → E → F → O

Visited: A



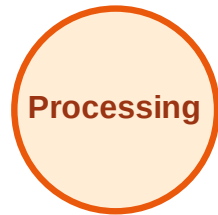
BFS Traversal

Step 5: Discover K, L, M from B

Legend



Complete



Processing



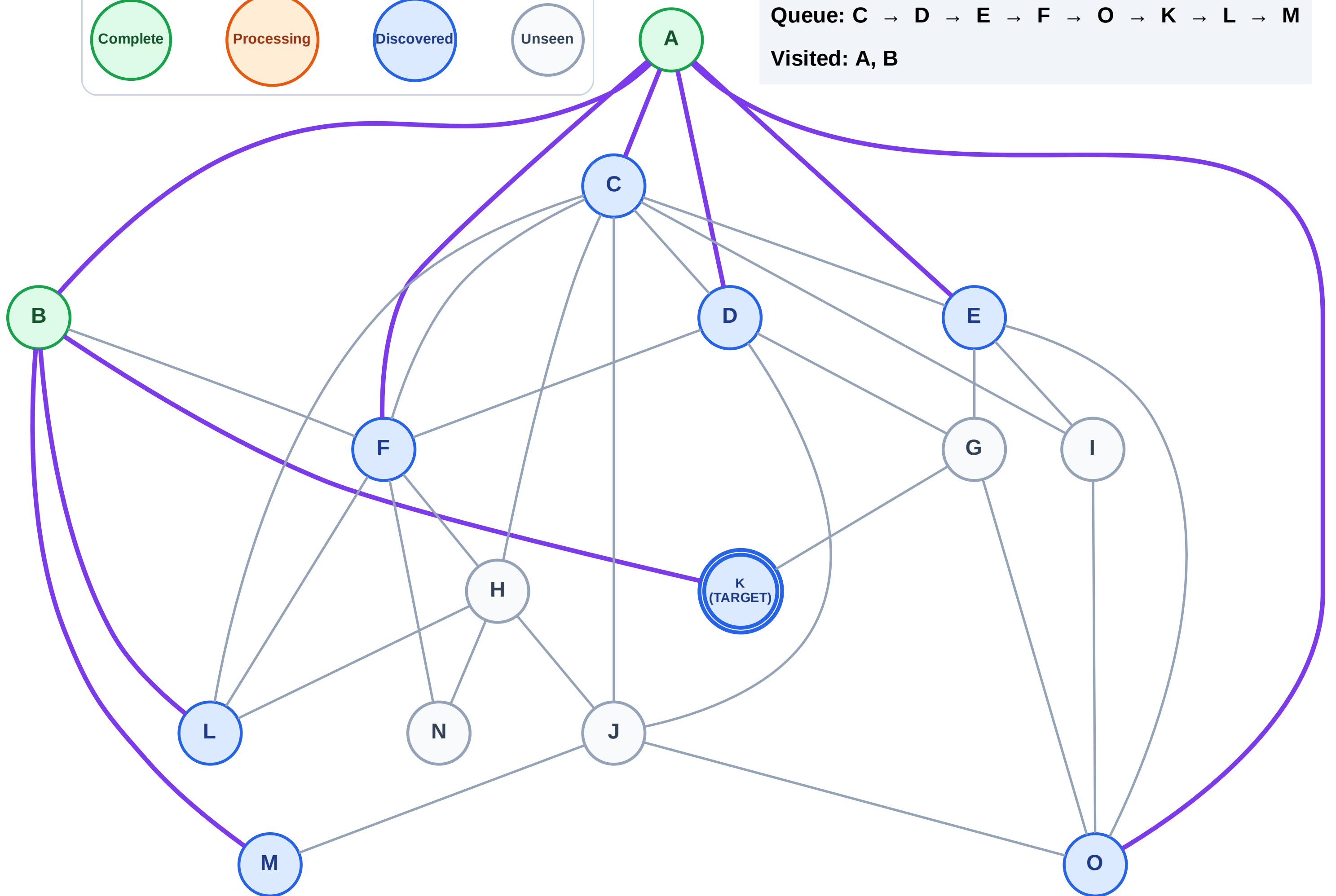
Discovered



Unseen

Queue: C → D → E → F → O → K → L → M

Visited: A, B



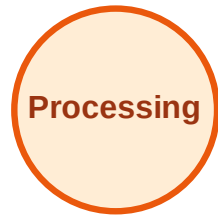
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



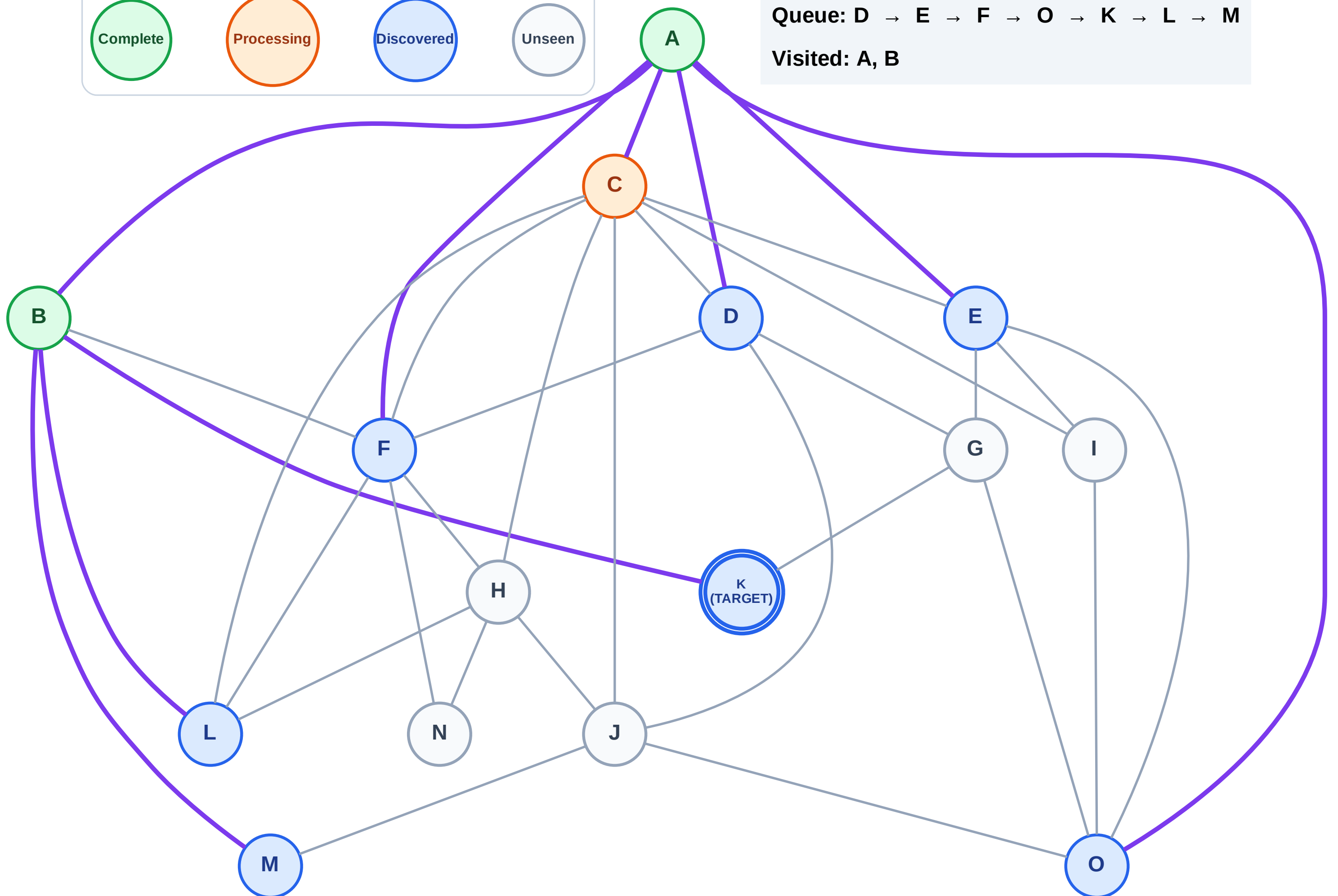
Discovered



Unseen

Queue: D → E → F → O → K → L → M

Visited: A, B



BFS Traversal

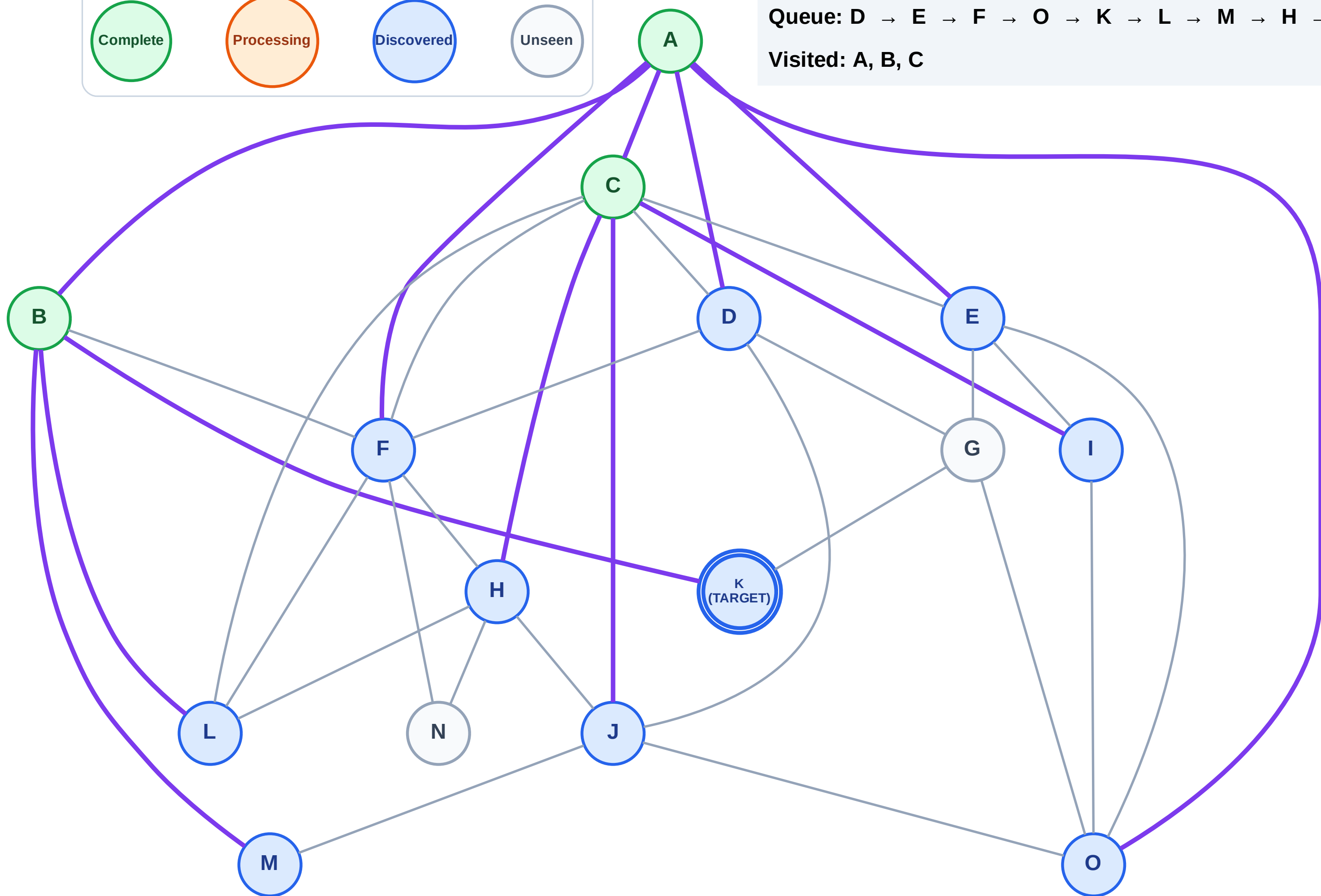
Step 7: Discover H, I, J from C

Legend



Queue: D → E → F → O → K → L → M → H → I → J

Visited: A, B, C



BFS Traversal

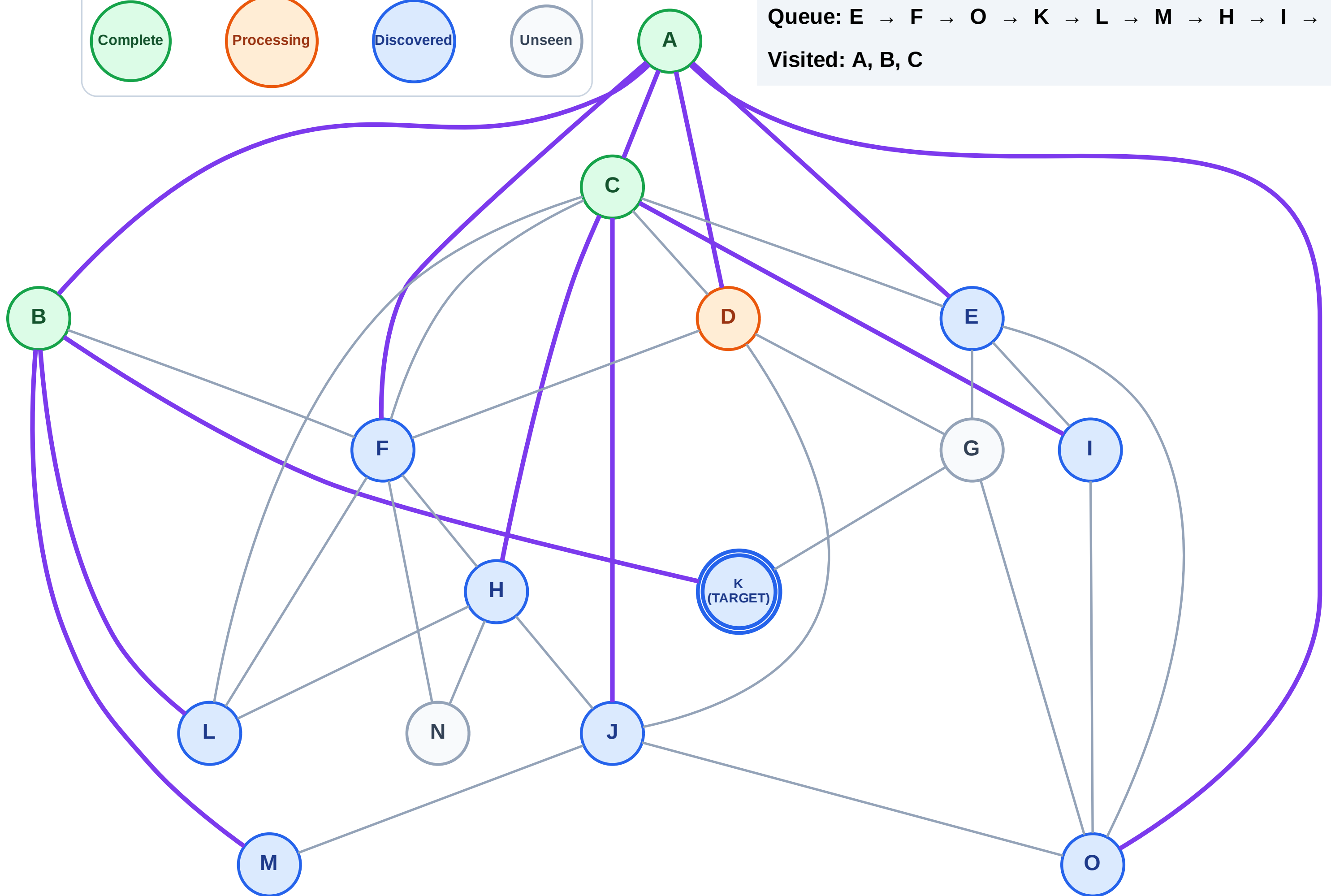
Step 8: Process node D

Legend



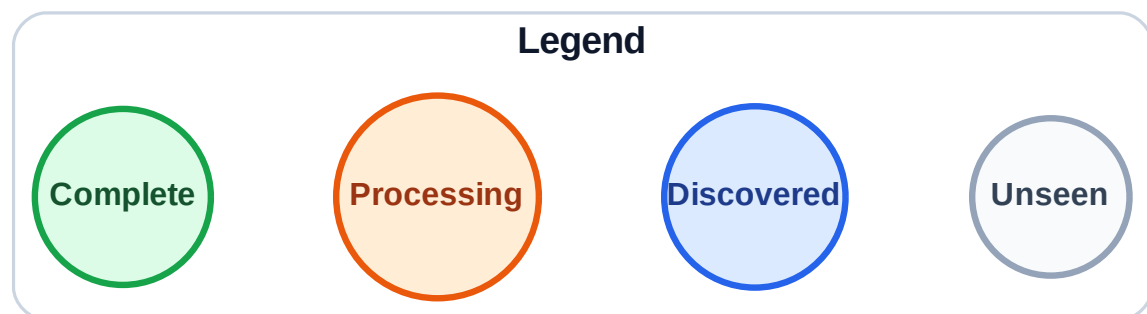
Queue: E → F → O → K → L → M → H → I → J

Visited: A, B, C



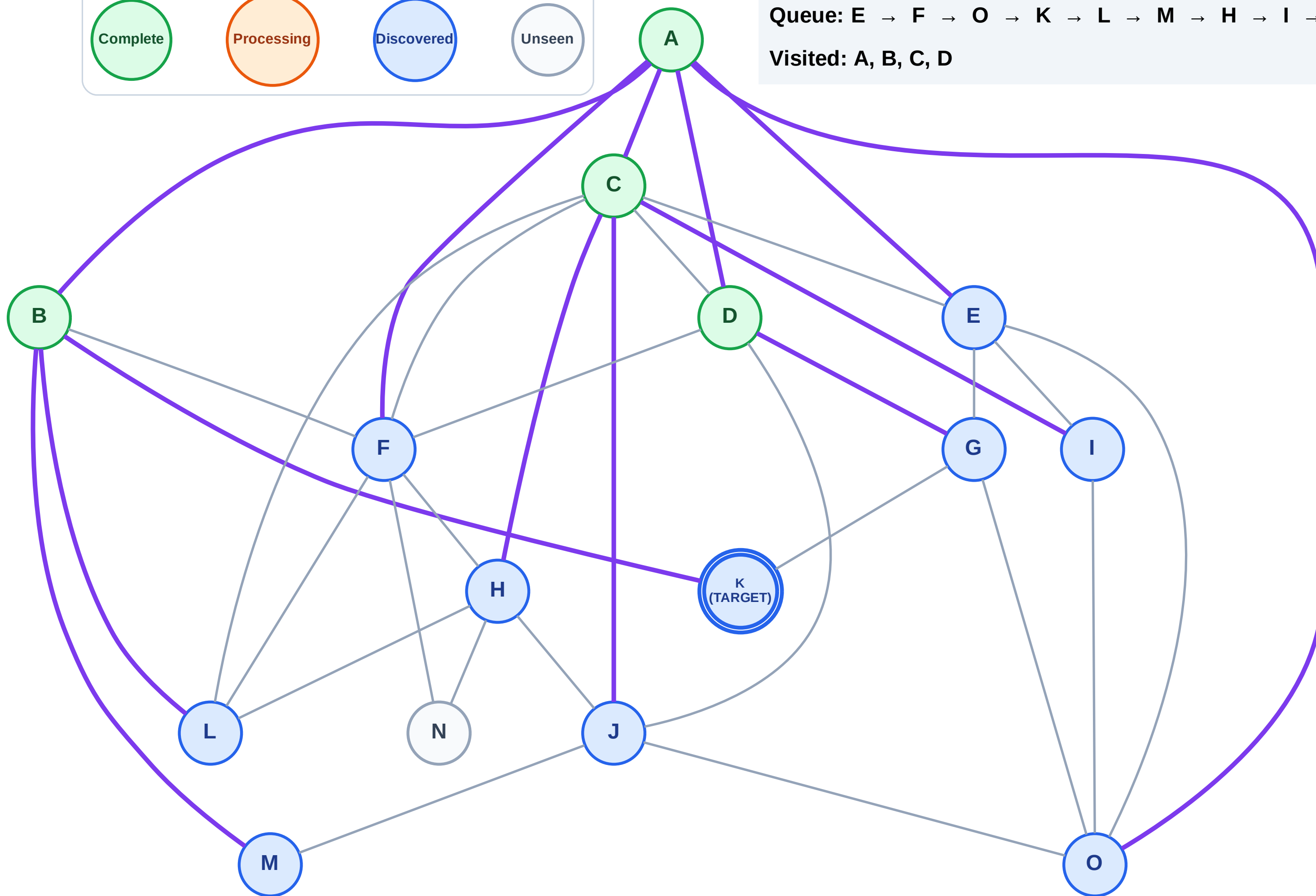
BFS Traversal

Step 9: Discover G from D



Queue: E → F → O → K → L → M → H → I → J → G

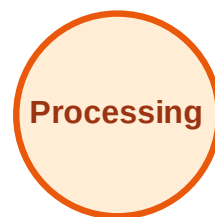
Visited: A, B, C, D



BFS Traversal

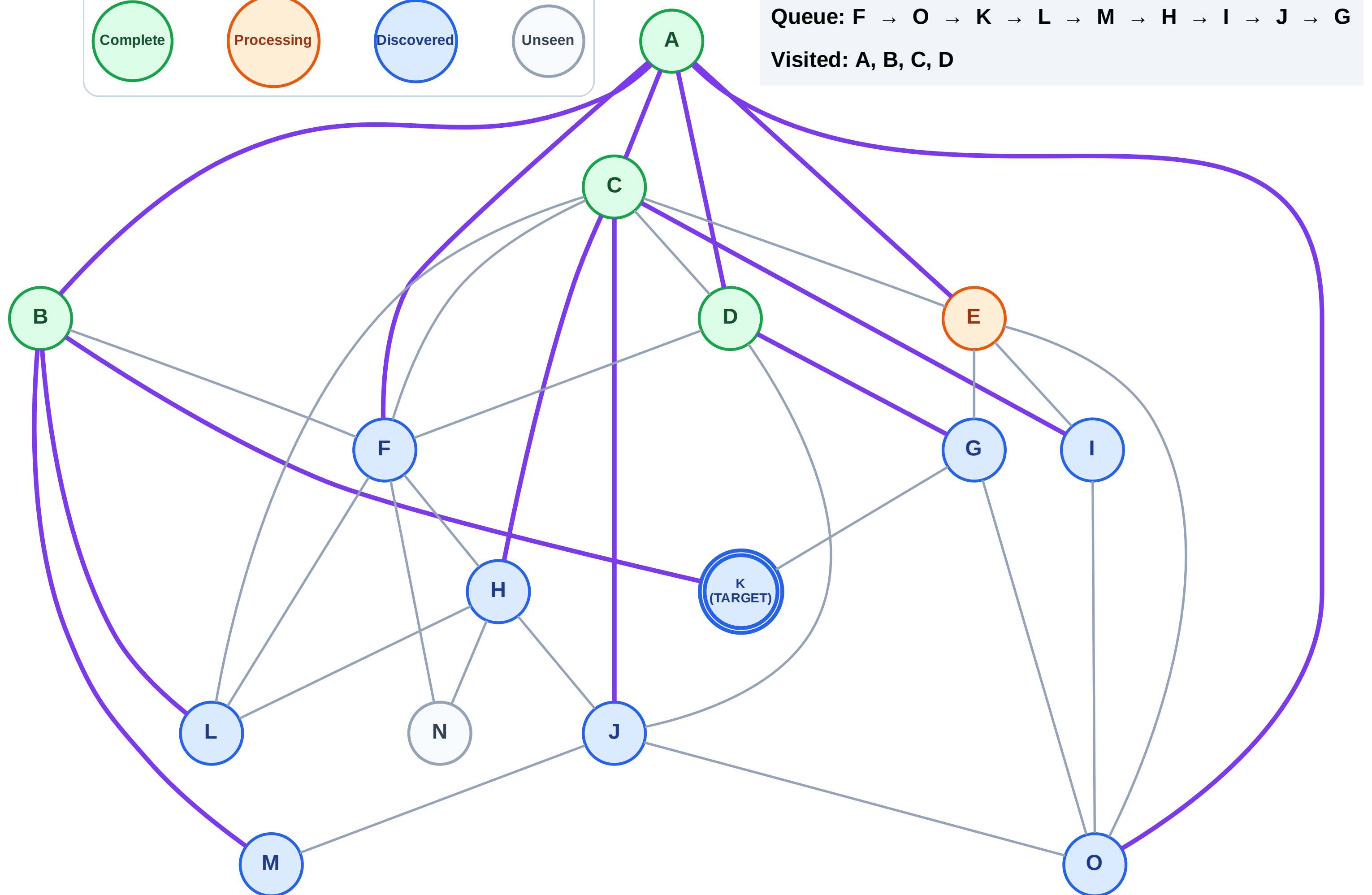
Step 10: Process node E

Legend



Queue: F → O → K → L → M → H → I → J → G

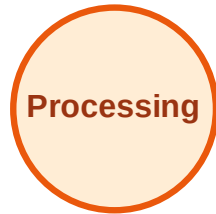
Visited: A, B, C, D



BFS Traversal

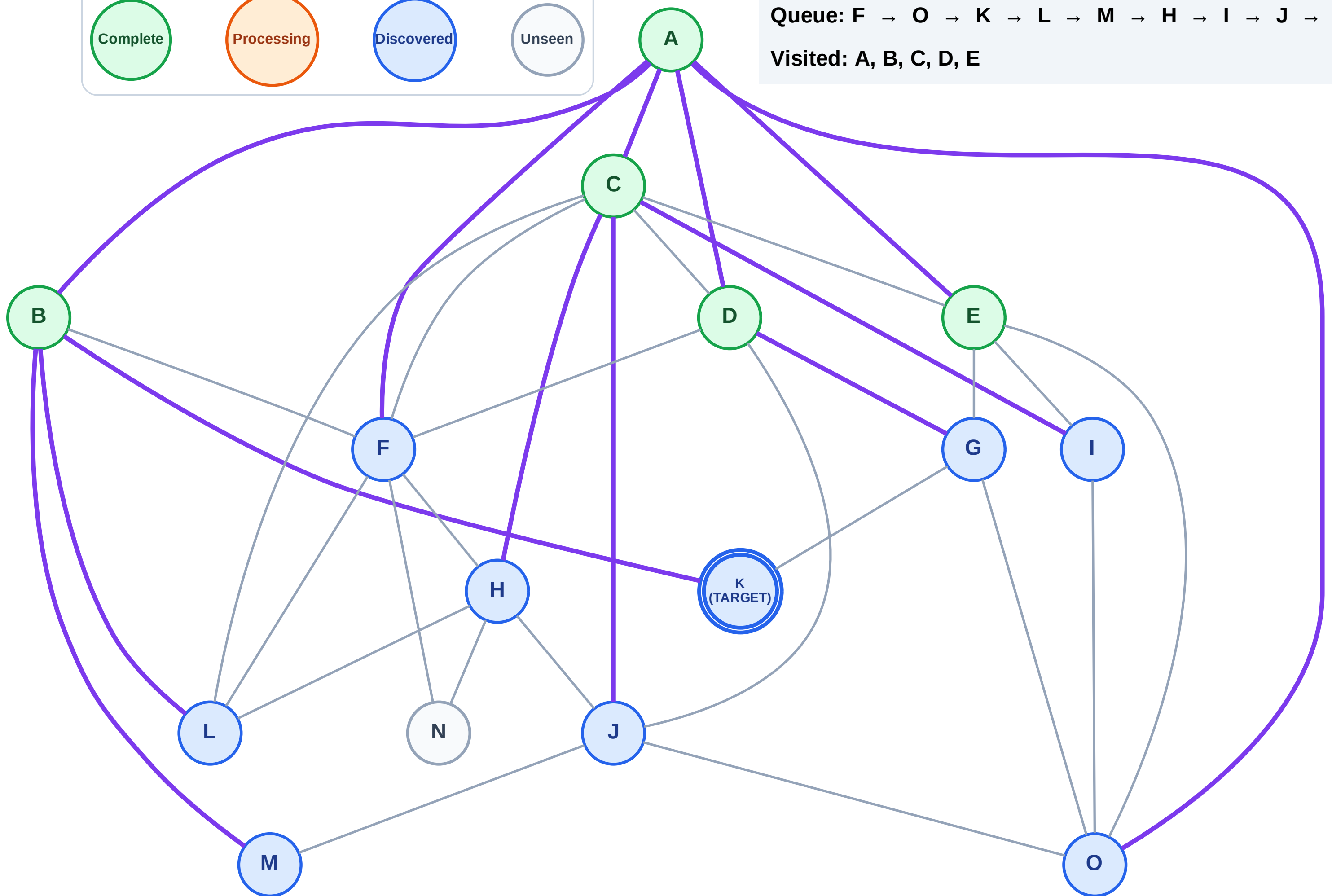
Step 11: No new neighbors from E

Legend



Queue: F → O → K → L → M → H → I → J → G

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node F

Legend

Complete

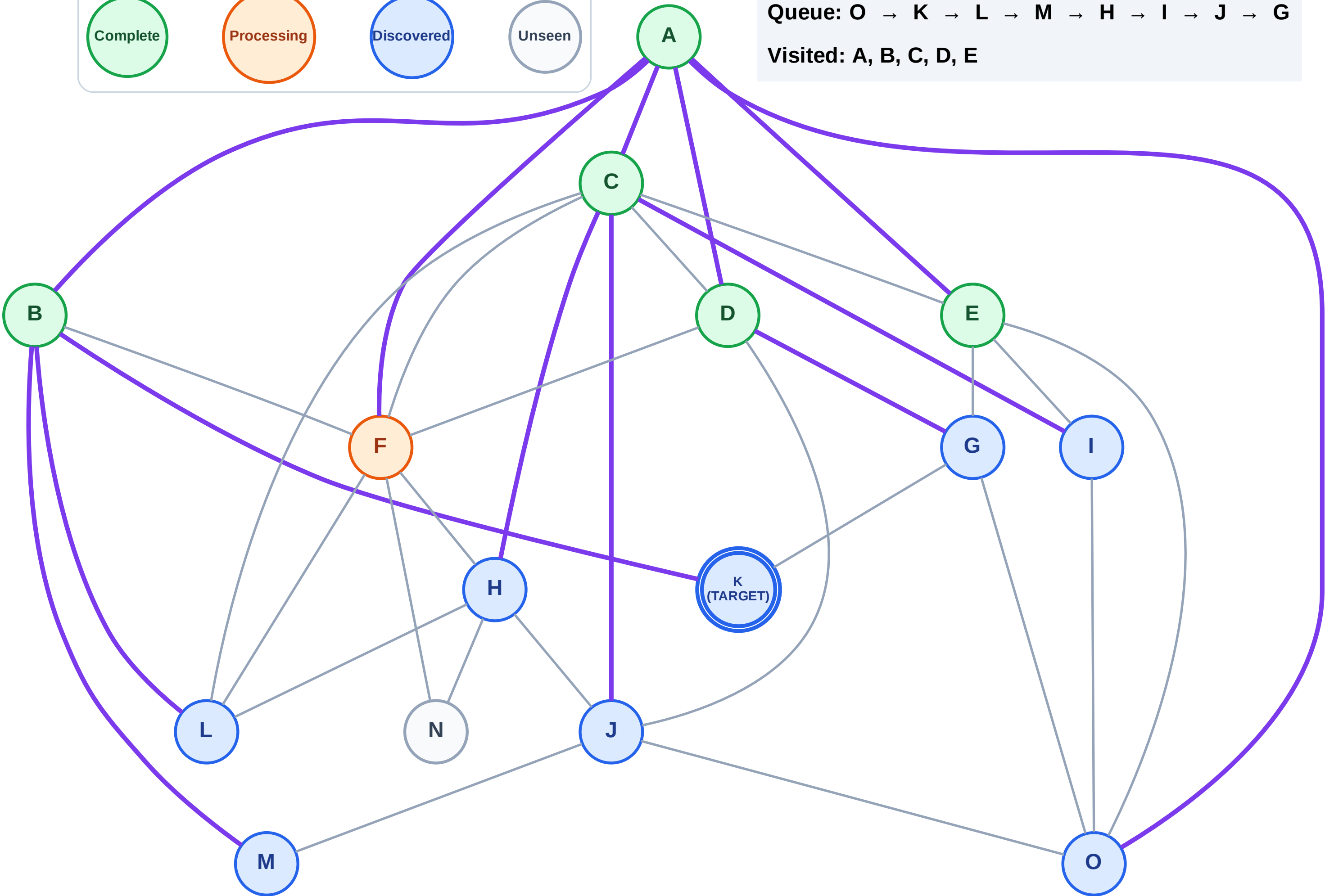
Processing

Discovered

Unseen

Queue: O → K → L → M → H → I → J → G

Visited: A, B, C, D, E



BFS Traversal

Step 13: Discover N from F

Legend

Complete

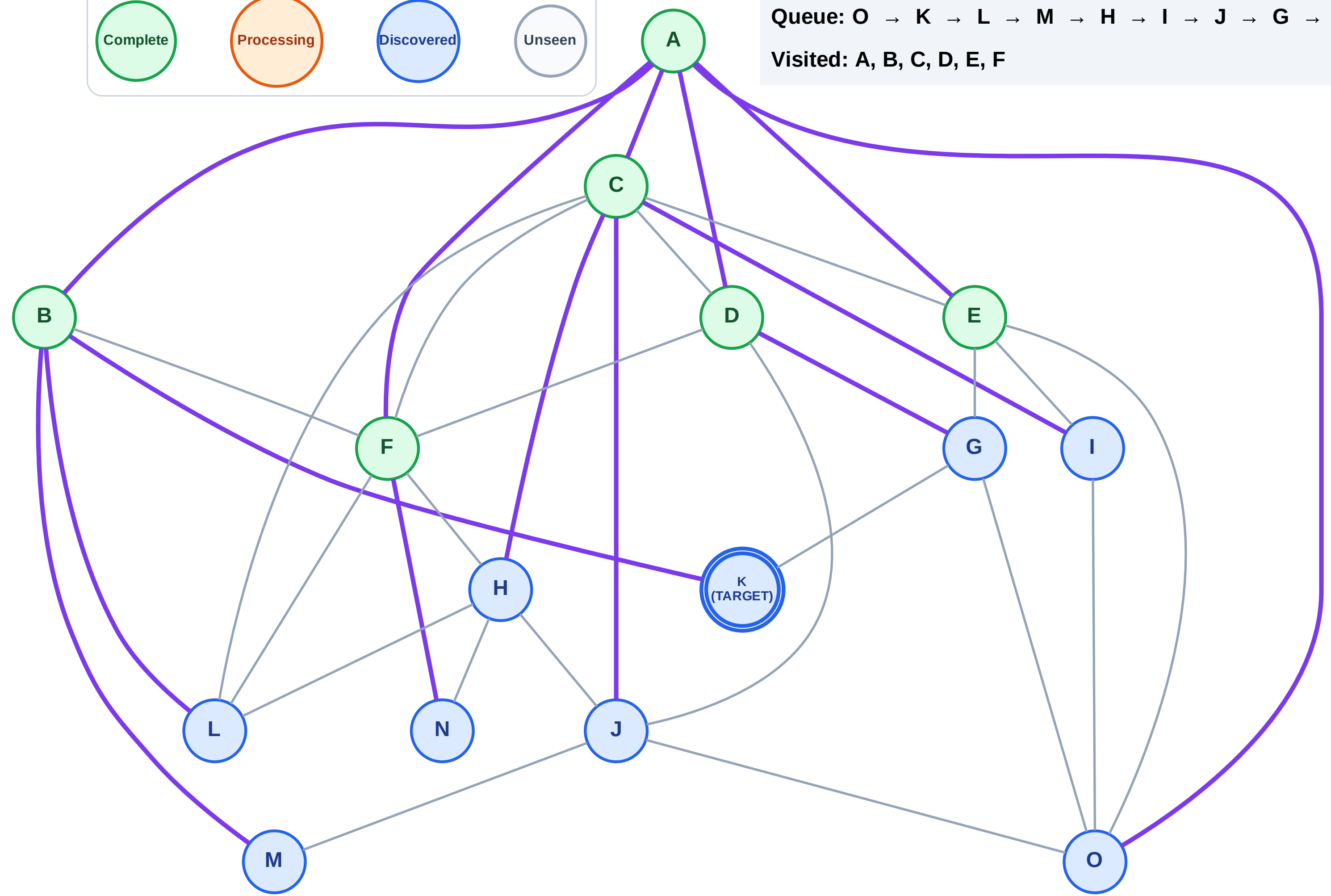
Processing

Discovered

Unseen

Queue: O → K → L → M → H → I → J → G → N

Visited: A, B, C, D, E, F



BFS Traversal

Step 14: Process node O

Legend

Complete

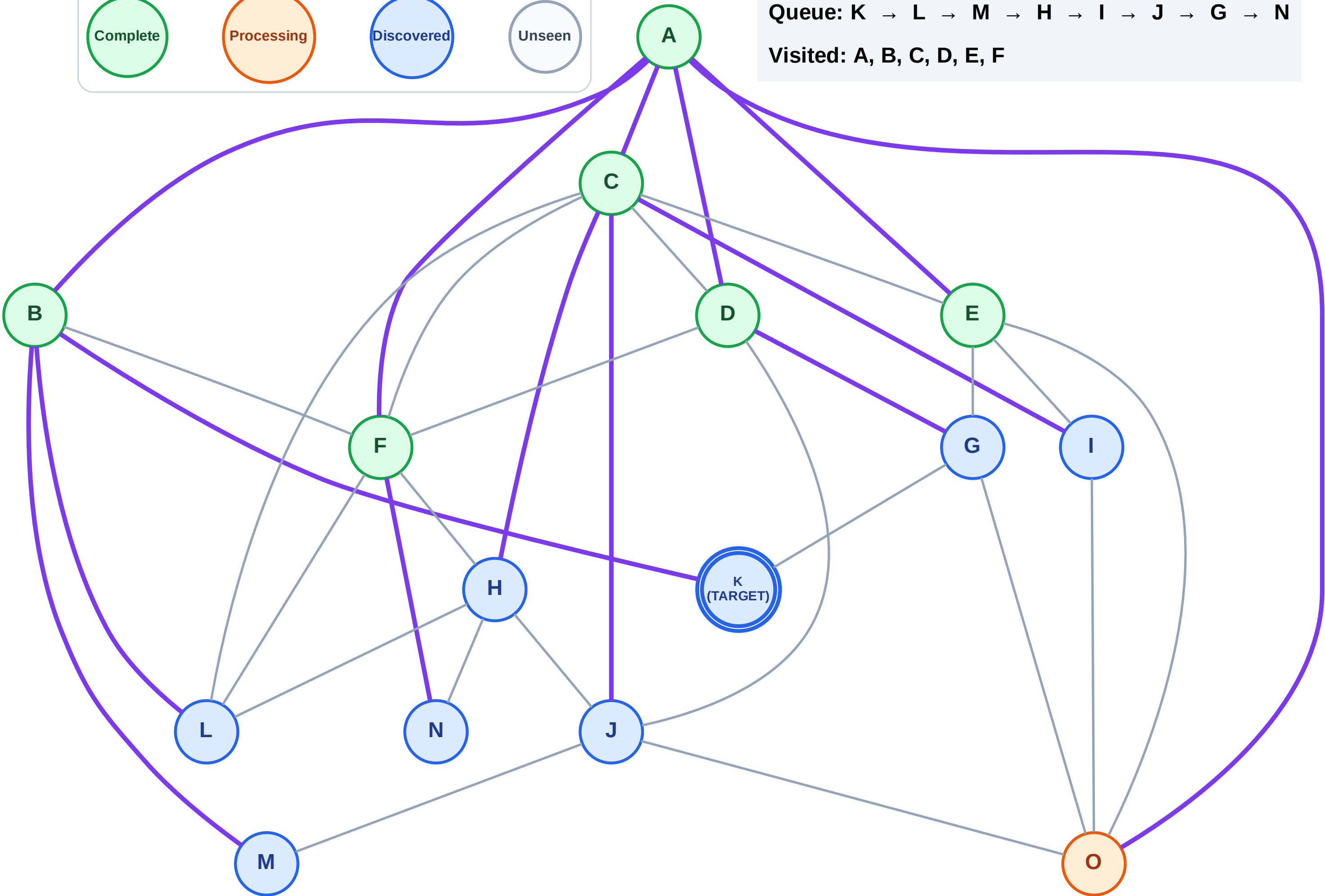
Processing

Discovered

Unseen

Queue: K → L → M → H → I → J → G → N

Visited: A, B, C, D, E, F



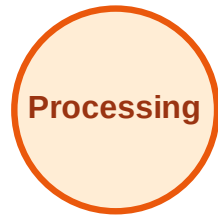
BFS Traversal

Step 15: No new neighbors from O

Legend



Complete



Processing



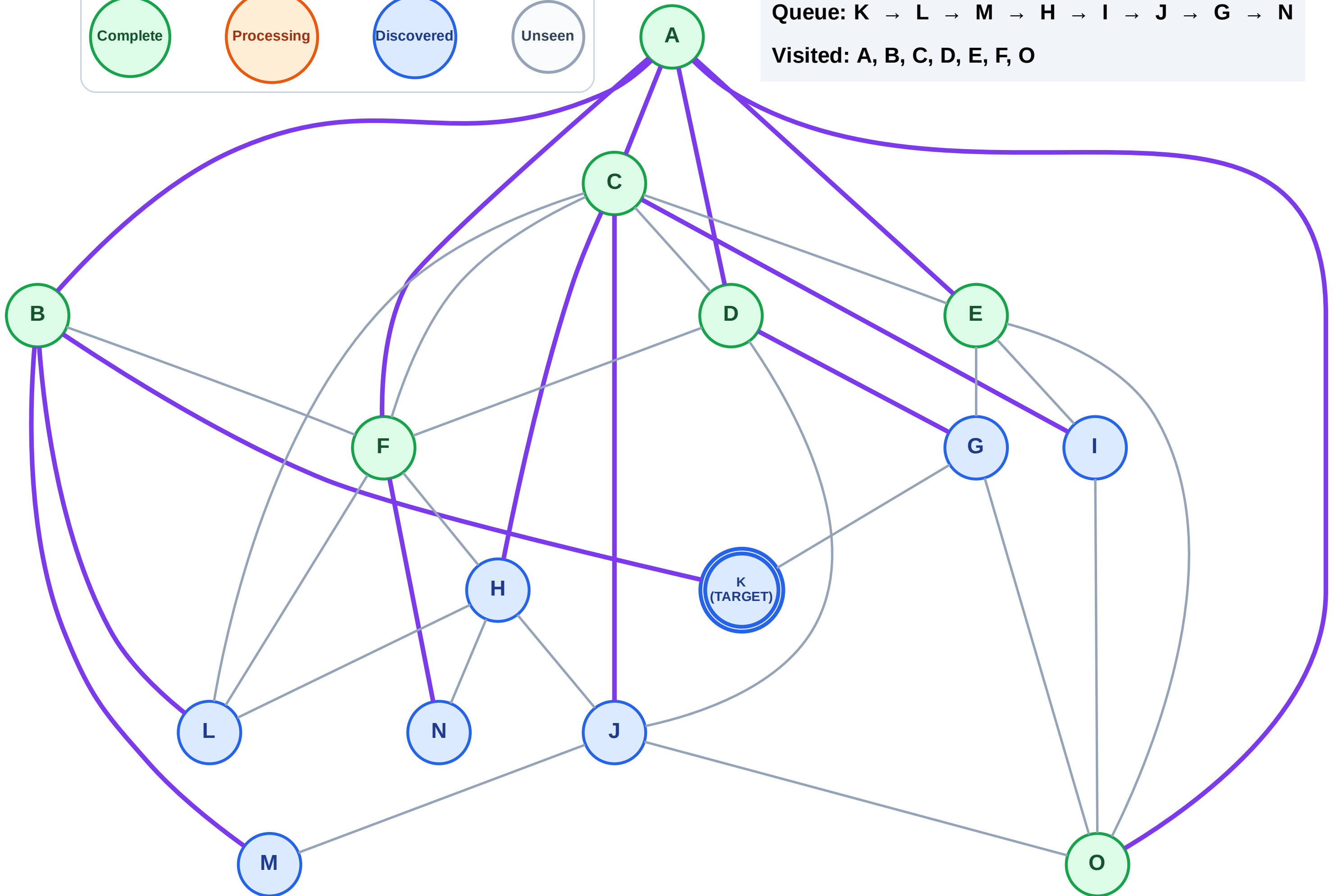
Discovered



Unseen

Queue: K → L → M → H → I → J → G → N

Visited: A, B, C, D, E, F, O



BFS Traversal

Step 16: Process node K

Legend

Complete

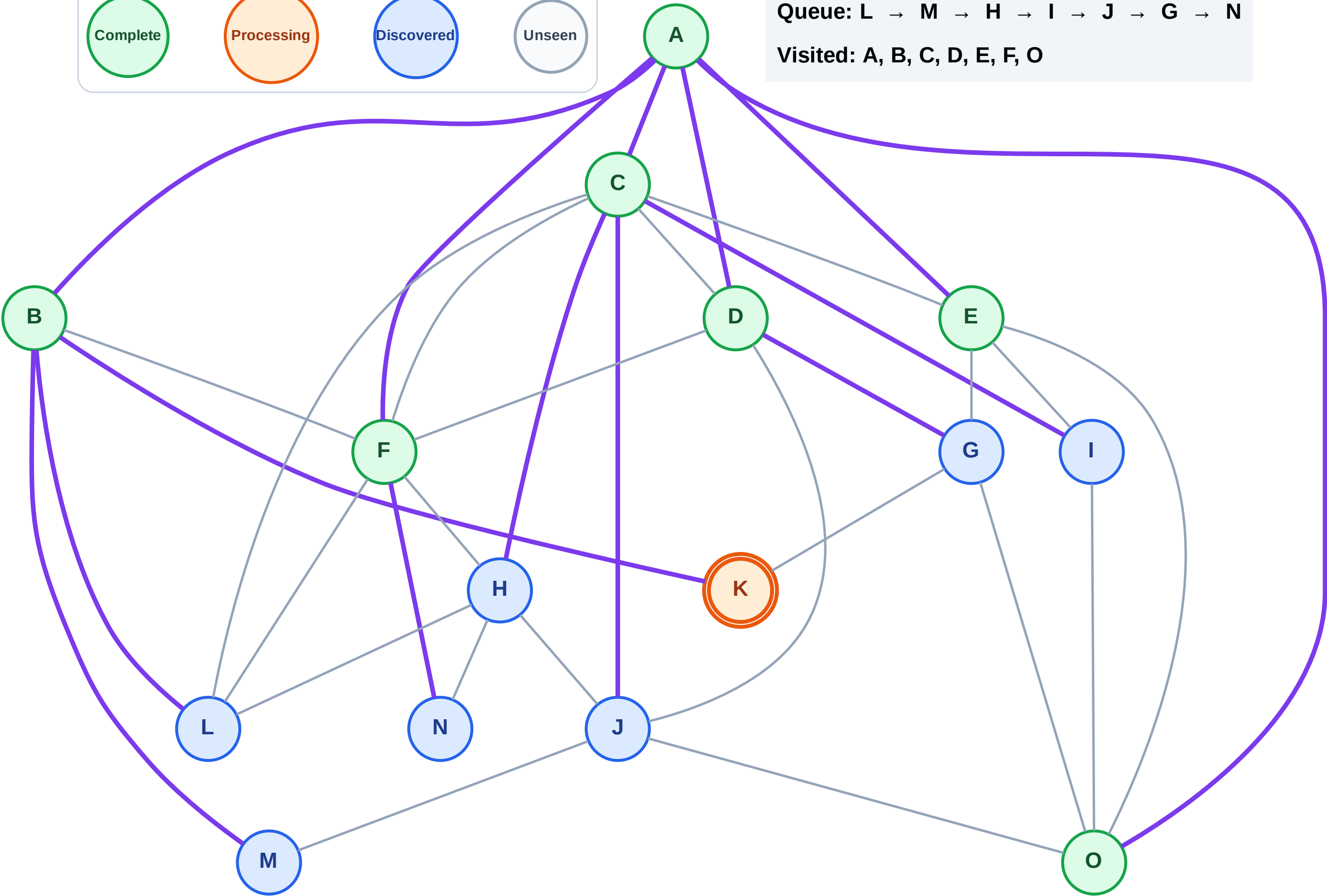
Processing

Discovered

Unseen

Queue: L → M → H → I → J → G → N

Visited: A, B, C, D, E, F, O



BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

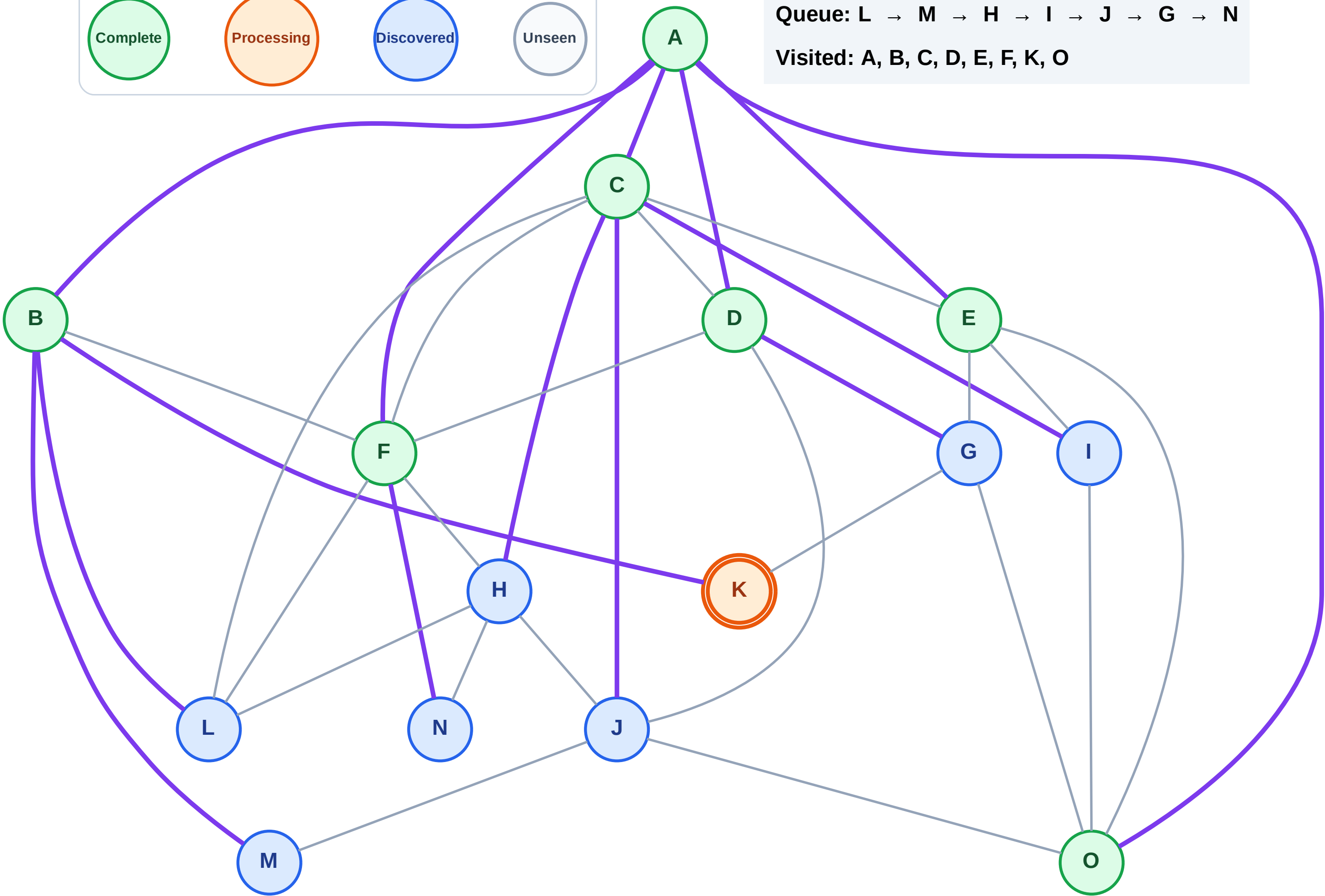
Processing

Discovered

Unseen

Queue: L → M → H → I → J → G → N

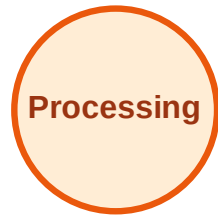
Visited: A, B, C, D, E, F, K, O



DFS Traversal

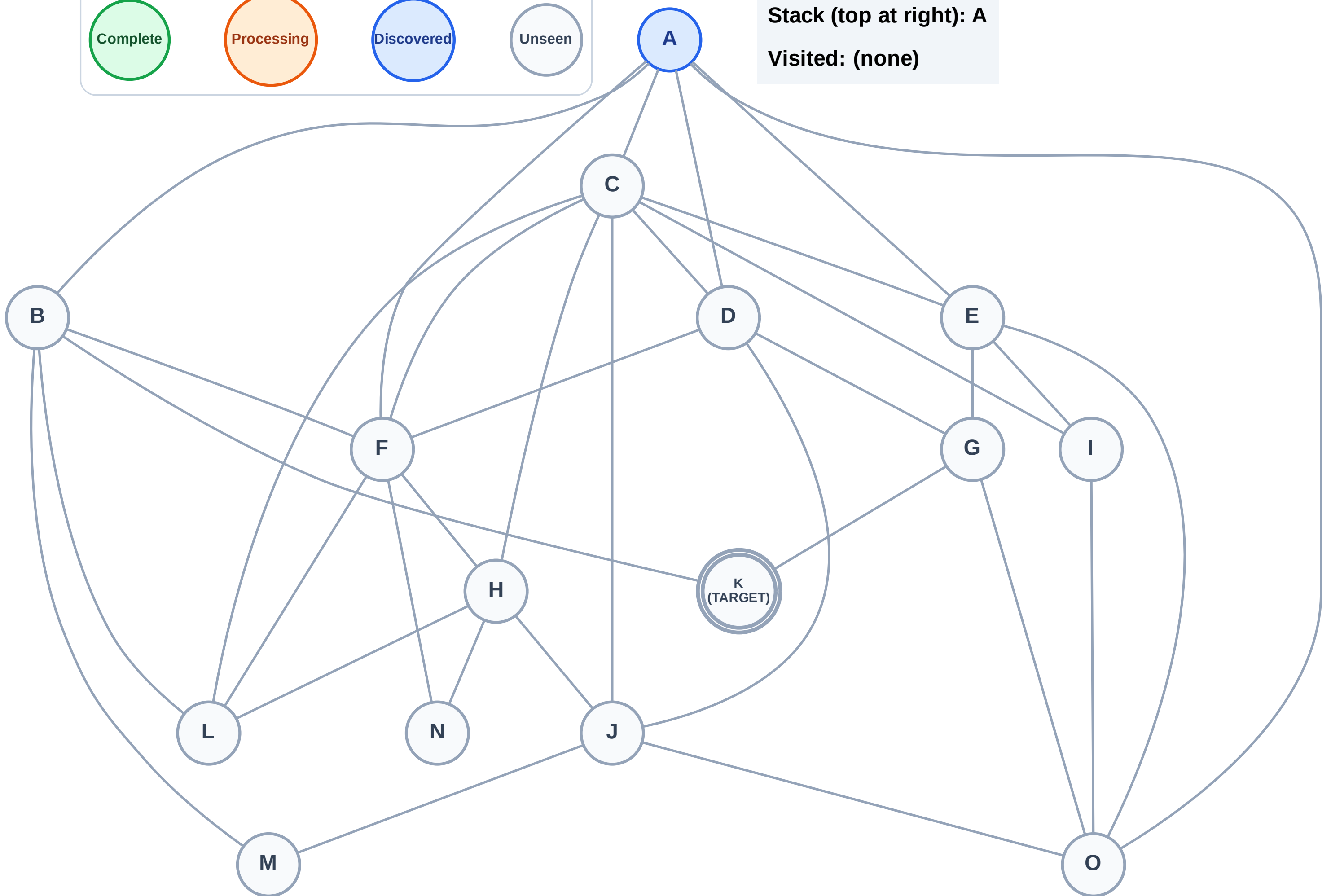
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

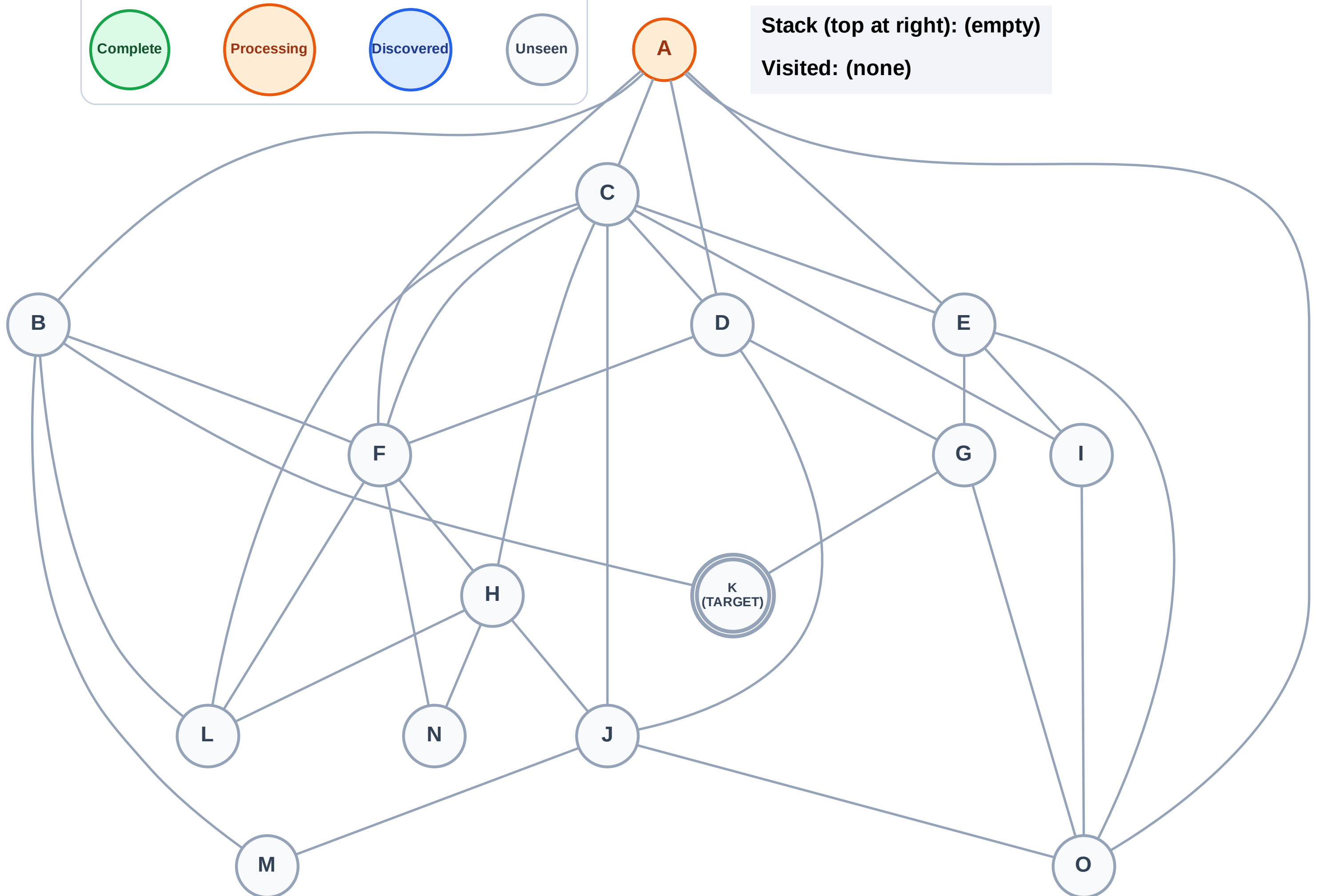
Step 2: Process node A

Legend



Stack (top at right): (empty)

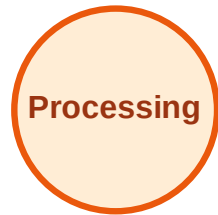
Visited: (none)



DFS Traversal

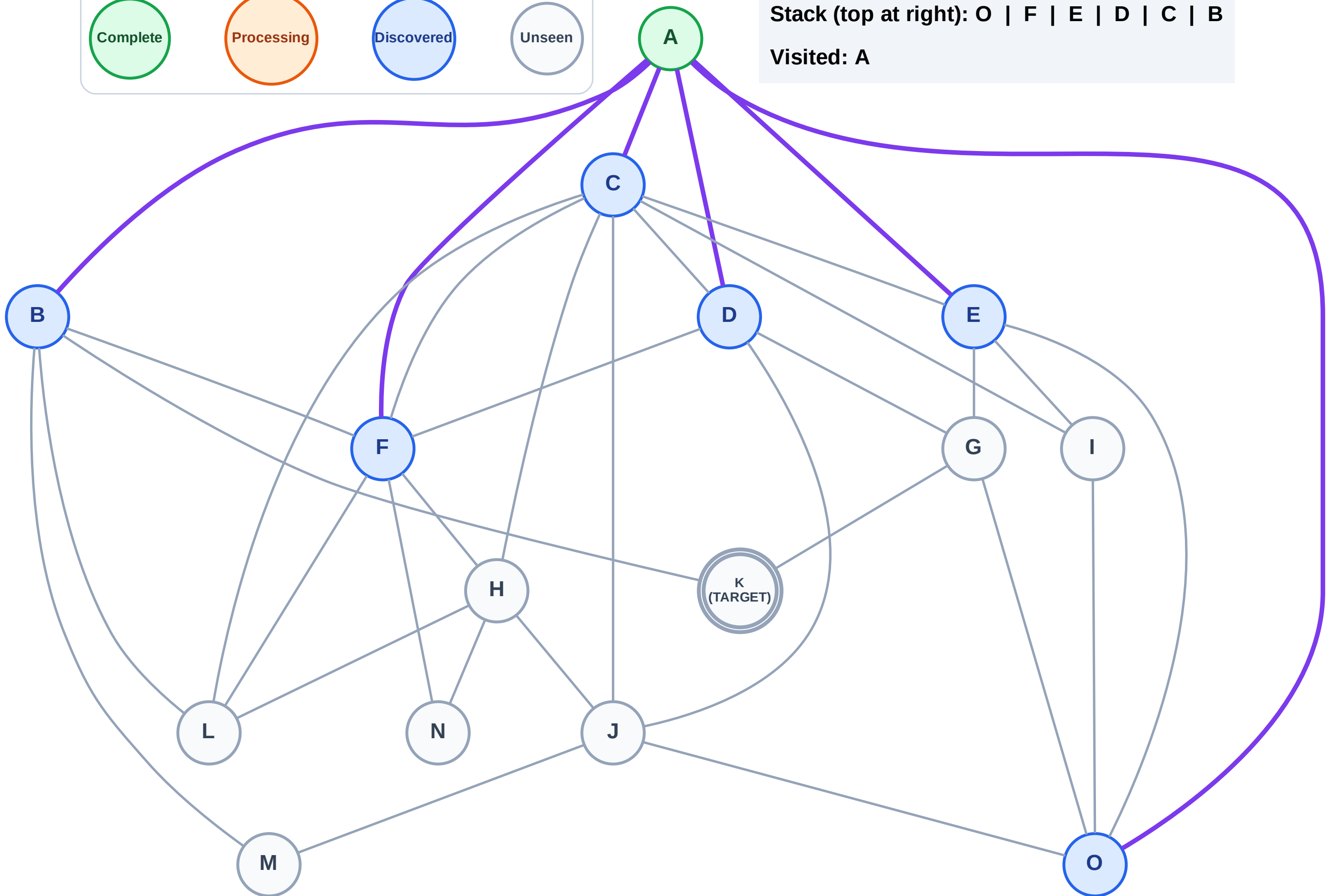
Step 3: Discover O, F, E, D, C, B from A

Legend



Stack (top at right): O | F | E | D | C | B

Visited: A



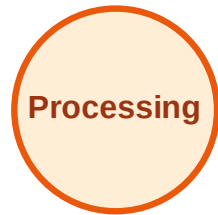
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



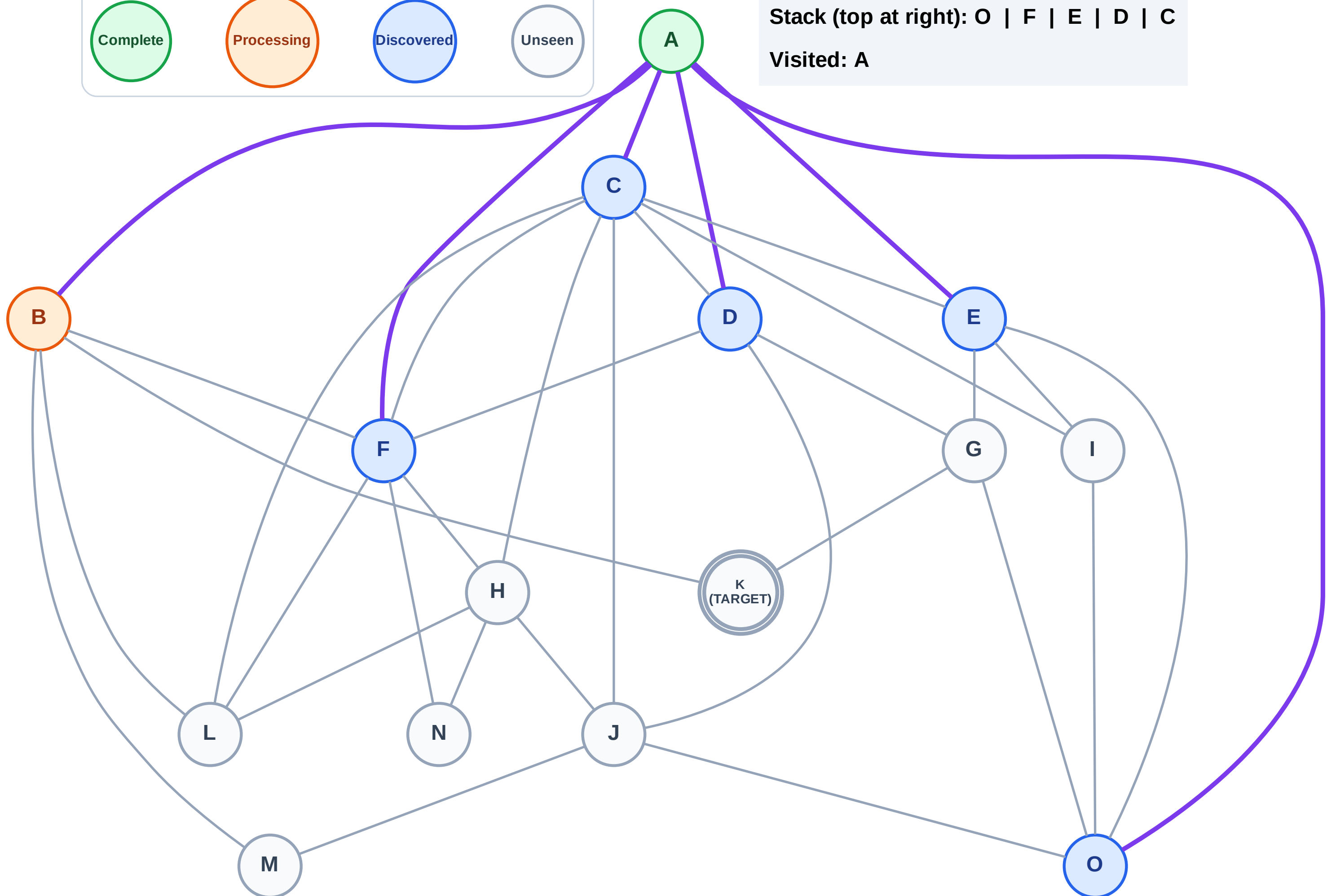
Discovered



Unseen

Stack (top at right): O | F | E | D | C

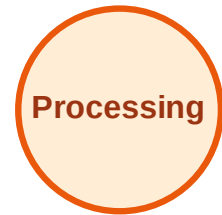
Visited: A



DFS Traversal

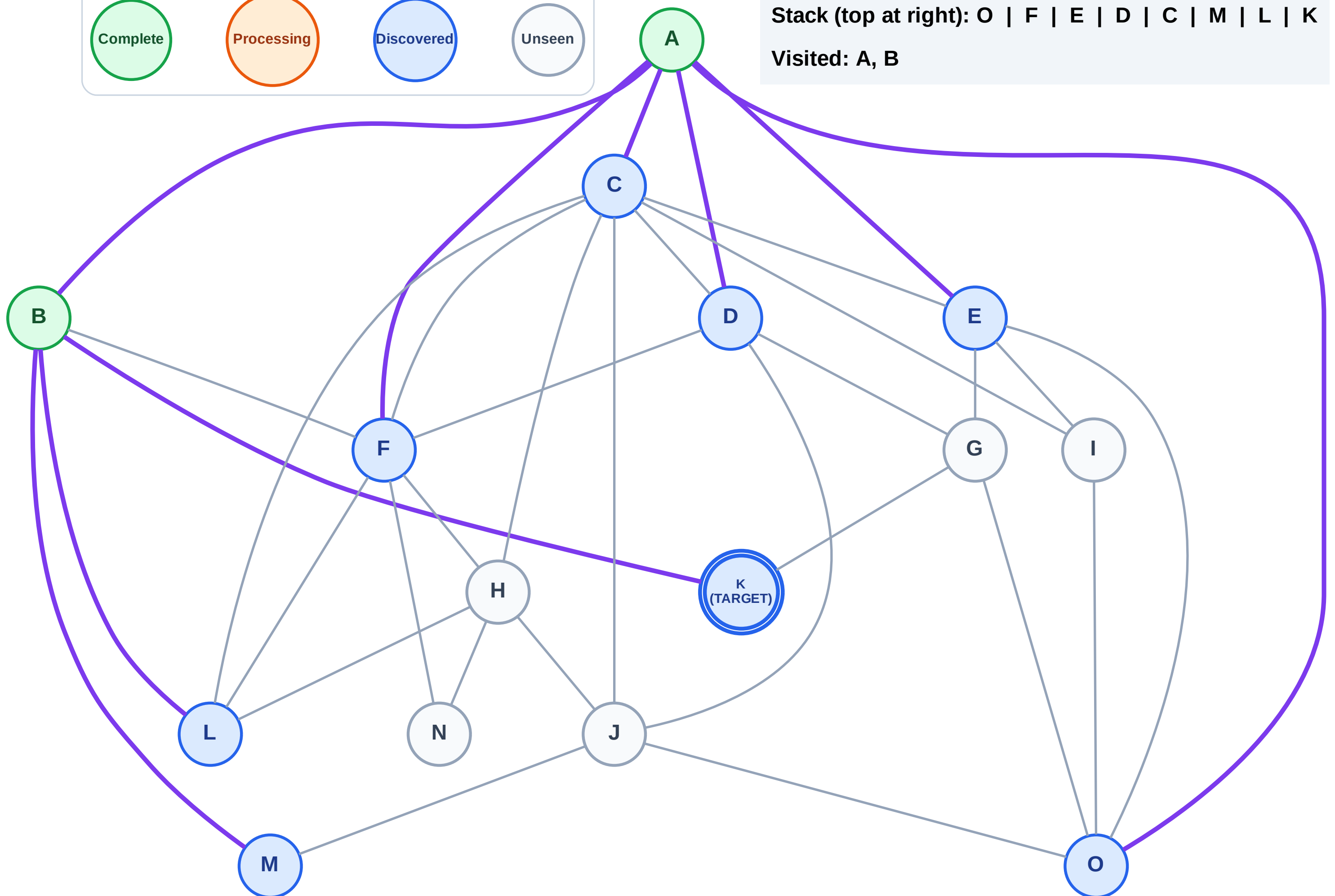
Step 5: Discover M, L, K from B

Legend



Stack (top at right): O | F | E | D | C | M | L | K

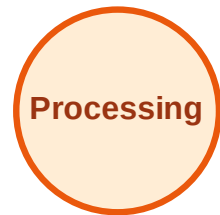
Visited: A, B



DFS Traversal

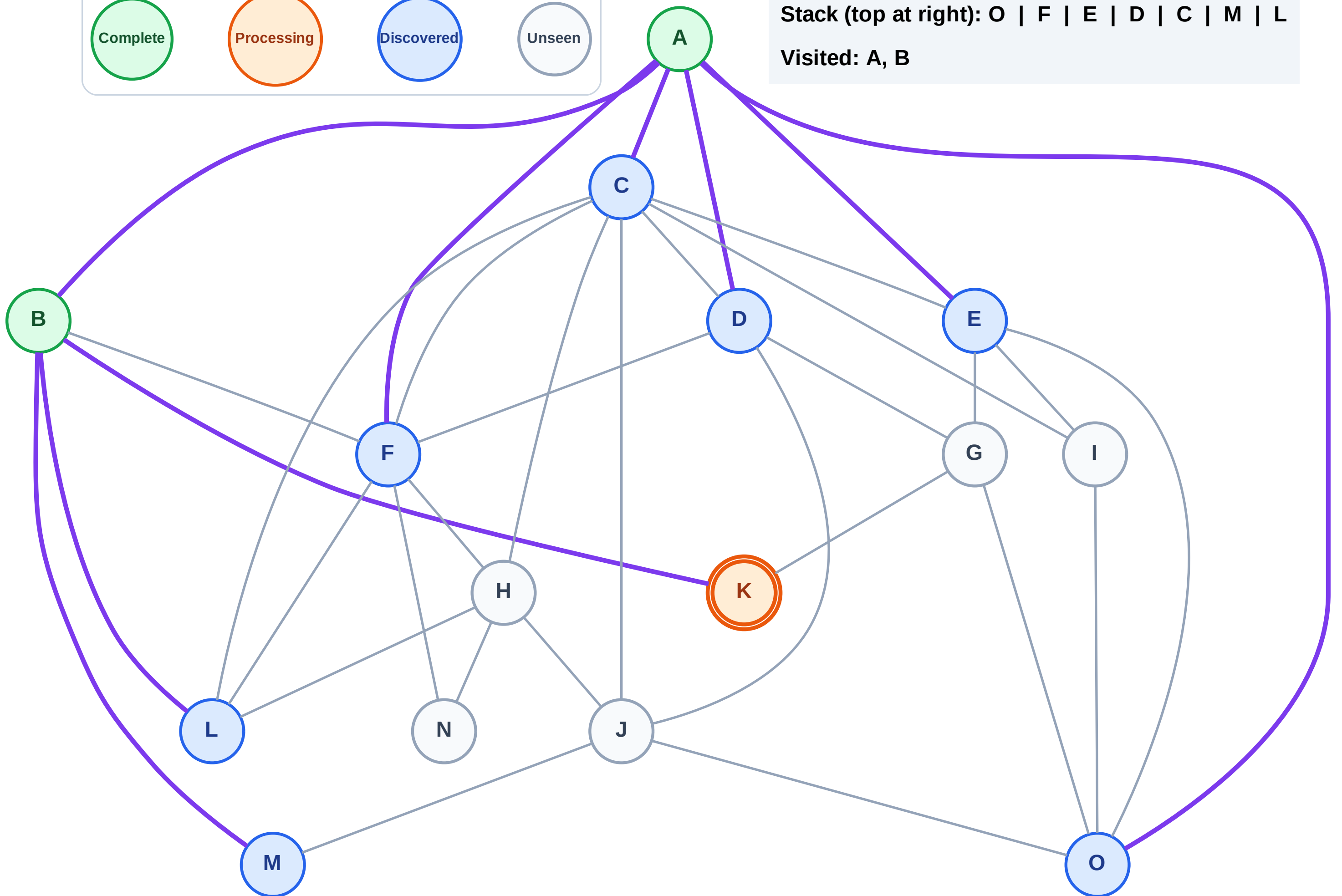
Step 6: Process node K

Legend



Stack (top at right): O | F | E | D | C | M | L

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | F | E | D | C | M | L

Visited: A, B, K

